US Patent & Trademark Office
Patent Public Search | Text View

United States Patent 12401366
Kind Code B2
Date of Patent August 26, 2025
Inventor(s) Sato; Hiroki et al.

Sensor interface circuit and sensor module

Abstract

A sensor interface circuit includes a frequency synchronization circuit connectable to a sensor. The frequency synchronization circuit includes: a reference voltage source generating a reference voltage; a current source connected to the reference voltage source, and generating a current by using the reference voltage; a voltage difference detection circuit including first and second input nodes and an output node, generating a control voltage according to a difference between voltages received at the first and second input nodes, one of the voltages received at the first and second input nodes corresponding to a detection value of the sensor; a voltage-controlled oscillation circuit connected to the output node, and generating an oscillation signal according to the control voltage; and a frequency/impedance conversion circuit connected between the voltage-controlled oscillation circuit and the second input node, and converting a frequency of a signal according to the oscillation signal into impedance.

Inventors: **Sato; Hiroki (Tokyo, JP), Sato; Masayuki (Tokyo, JP), Shirakawa; Takaaki (Tokyo, JP), Pan; Chunhui (Tokyo, JP), Ishihara; Noboru (Tokyo, JP), Murata; Shinji (Tokyo, JP)**

Applicant: **NISSHINBO MICRO DEVICES INC.** (Tokyo, JP); **TOKYO INSTITUTE OF TECHNOLOGY** (Tokyo, JP); **ALPS ALPINE CO., LTD.** (Tokyo, JP)

Family ID: **1000008778256**

Assignee: **NISSHINBO MICRO DEVICES INC. (Tokyo, JP); TOKYO INSTITUTE OF TECHNOLOGY (Tokyo, JP); ALPS ALPINE CO., LTD. (Tokyo, JP)**

Appl. No.: **18/286656**

Filed (or PCT Filed): **May 26, 2021**

PCT No.: **PCT/JP2021/020031**

PCT Pub. No.: **WO2022/249346**

PCT Pub. Date: December 01, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240195422 A1	Jun. 13, 2024

Publication Classification

Int. Cl.: **H03L7/093** (20060101); **G01R27/02** (20060101); **H03K17/687** (20060101)

U.S. Cl.:

CPC **H03L7/093** (20130101); **H03K17/6871** (20130101); G01R27/02 (20130101)

Field of Classification Search

CPC: H03L (7/093); H03K (17/6871); G01R (27/02)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2010/0283513	12/2009	Horng	327/105	H03D 3/001
2011/0009083	12/2010	Namba	455/318	H03L 7/1976
2016/0099682	12/2015	Liu	331/116FE	H03B 5/364
2020/0099383	12/2019	Kubo	N/A	H03L 7/10

OTHER PUBLICATIONS

Miyauchi, Kaede et al. “Low-Power Wireless Sensor Module for RF Backscattering”. IEICE General Conference 2018, Japan, Mar. 20-23, 2018, B-18-17, p. 361. cited by applicant
Jul. 6, 2021 International Search Report issued in International Patent Application No. PCT/JP2021/020031. cited by applicant

Primary Examiner: Rodak; Lee E

Assistant Examiner: Ferdous; Zannatul

Attorney, Agent or Firm: Oliff PLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application is national stage application of International Application No. PCT/JP2021/020031, filed on May 26, 2021, which designates the United States, the entire

contents of which are incorporated herein by reference.
TECHNICAL FIELD
(2) Embodiments described herein relate generally to a sensor interface circuit and a sensor module.
(3) A sensor interface circuit having an oscillation circuit oscillates the oscillation circuit according to a level of a signal of a sensor, and generates and outputs an oscillation signal having a frequency according to the signal of the sensor. At this time, the sensor interface circuit is required to convert the signal of the sensor into frequency with high accuracy (e.g., Non-patent Literature 1: Kaede Miyauchi, Taichi Taguchi, Yosuke Ishikawa, Hiroyuki Ito, Shiro Dosho, Kazuya Masu, and Noboru Ishihara, "Evaluation Result of Test Production of Low-Power Wireless Sensor Terminal Module for RF Backscattering", 2018 IEICE General Conference, Japan, Mar. 20-23, 2018, B-18-17, p. 361).
(4) For example, in an interface circuit, when an original oscillator such as crystal is used to convert a signal of a sensor into frequency with high accuracy, a current according to an oscillation signal of the original oscillator may continue to flow to increase current consumption; hence, it is desired to reduce current consumption.
(5) The present invention has been made in view of the above, and an object of the present invention is to provide a sensor interface circuit and a sensor module capable of reducing current consumption.
SUMMARY
(6) To solve the problem described above and achieve the object, a sensor interface circuit according to one aspect of the present invention includes a frequency synchronization circuit connectable to a sensor. The frequency synchronization circuit includes: a reference voltage source configured to generate a reference voltage; a current source connected to the reference voltage source, the current source being configured to generate a current by using the reference voltage; a voltage difference detection circuit including a first input node connected to the reference voltage source, a second input node connected to the current source, and an output node, one of a voltage received at the first input node and a voltage received at the second input node corresponding to a detection value of the sensor, and the voltage difference detection circuit being configured to generate a control voltage according to a difference between the voltage received at the first input node and the voltage received at the second input node; a voltage-controlled oscillation circuit connected to the output node of the voltage difference detection circuit, the voltage-controlled oscillation circuit being configured to generate an oscillation signal according to the control voltage; and a frequency/impedance conversion circuit connected between the voltage-controlled oscillation circuit and the second input node of the voltage difference detection circuit, the frequency/impedance conversion circuit being configured to convert a frequency of a signal according to the oscillation signal into impedance.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a diagram illustrating a configuration of a communication system to which a sensor module including a sensor interface circuit according to a first embodiment is applied;
(2) FIG. **2** is a diagram illustrating a configuration of the sensor module including the sensor interface circuit according to the first embodiment;
(3) FIG. **3** is a diagram illustrating a configuration of a frequency synchronization circuit in the first embodiment;
(4) FIG. **4** is a diagram illustrating an operation of a frequency/impedance conversion circuit in the first embodiment;

(5) FIG. **5** is a waveform diagram illustrating an operation of the frequency synchronization circuit in the first embodiment;
(6) FIG. **6** is a diagram illustrating a configuration of a sensor module including a sensor interface circuit according to a second embodiment;
(7) FIG. **7** is a diagram illustrating a configuration of a frequency synchronization circuit in the second embodiment; and
(8) FIG. **8** is a waveform diagram illustrating an operation of the frequency synchronization circuit in the second embodiment.

DETAILED DESCRIPTION

(9) Hereinafter, embodiments of a sensor interface circuit are described in detail with reference to the drawings. In the following embodiments, portions denoted by the same reference numeral perform similar operations, and a repeated description is omitted as appropriate.

First Embodiment

(10) A sensor interface circuit according to a first embodiment is a circuit that converts a signal of a sensor into frequency. In the case where this circuit is used for Internet of Things (IoT) technology, it is conceivable to configure the circuit such that a signal according to a frequency converted by the sensor interface circuit can be communicated by RF backscatter communication.

(11) For example, a sensor module **100** including a sensor interface circuit **1** is configured as illustrated in FIG. **1**. FIG. **1** is a diagram illustrating a configuration of a communication system **300** including a sensor module **100** to which a sensor interface circuit **1** is applied.

(12) In the communication system **300**, a plurality of sensor modules **100-1** to **100-***n* and an information collection terminal **200** are configured to be capable of RF backscatter communication. n is an integer of 1 or more. The information collection terminal **200** can transmit an RF signal to each sensor module **100**. Each sensor module **100** can, by using the RF signal, transmit a signal according to a detection value of a sensor to the information collection terminal **200**.

(13) Each sensor module **100** includes a sensor **2**, the sensor interface circuit **1**, an impedance conversion circuit **4**, and an antenna **5**. The sensor interface circuit **1** is electrically connected between the sensor **2** and the antenna **5**. The sensor interface circuit **1** includes an oscillation circuit and an RF switch, and performs RF backscatter communication by, with a signal from the sensor **2**, changing the frequency of the oscillation circuit and using the oscillation signal to control the on/off of the RF switch. That is, the sensor interface circuit **1** changes the impedance on the RF switch side as viewed from the antenna **5** and causes an RF signal from the information collection terminal **200** to be reflected and absorbed according to sensor information, and transmits the resulting signal to the information collection terminal **200**. Thereby, in the sensor interface circuit **1**, a signal of sensor information can be transmitted to the information collection terminal **200** while current consumption is reduced.

(14) The oscillation frequency of the oscillation circuit is defined by, for example, any one of or a combination of an amplification circuit, a resistance, a capacitance, an inductor, and a delay element. When the oscillation circuit is a relaxation oscillation circuit configured with an inverter, a resistance R, and a capacitance C, the oscillation period T is theoretically expressed by Mathematical Formula 1 below.

$$T=2.\mathrm{Math}.\ln 3.\mathrm{Math}.(R.\mathrm{Math}.C)+Td \qquad (1)$$

(15) In Mathematical Formula 1, Td is the delay time of the inverter. To obtain high stability performance with respect to temperature with this relaxation oscillation circuit, it is desired to bring the temperature coefficients of the resistance R, the capacitance C, and the delay time Td close to zero. Since the delay time Td is the delay time of the inverter, it is difficult to obtain high stability performance with respect to temperature because the delay time Td greatly changes with power supply voltage fluctuation or temperature fluctuation.

(16) As a specific example in which temperature stability is a problem, there is an application that detects the amount of change in frequency. By connecting the sensor to the relaxation oscillation

circuit, for example, a change in oscillation frequency can be obtained according to a change in the resistance of the sensor. There is a correlation between the amount of change in the resistance of the sensor and the amount of change in oscillation frequency. For example, when the signal of the sensor changes by 1%, the frequency changes by 1%. When the relaxation oscillation circuit is oscillated at 100 kHz, the frequency change is 1 kHz; to detect the change of 1 kHz with, for example, 64 gradations, it is desired that oscillation frequency be generated with an accuracy corresponding to the amount of change of 1 kHz/64=15.625 Hz.
(17) In order to increase the accuracy of the oscillation frequency of the oscillation circuit, it is conceivable to create an oscillation circuit by using a frequency synchronization circuit. For example, the frequency synchronization circuit is a frequency negative feedback circuit configured with an original oscillator, a frequency comparison circuit, a frequency division circuit, and a voltage-controlled oscillation circuit (VCO). The frequency difference between the oscillation frequency of the voltage-controlled oscillation circuit and a reference oscillation frequency of the original oscillator is detected, and the voltage of the voltage-controlled oscillation circuit is controlled such that the frequency difference approaches zero. As a result, the voltage-controlled oscillation circuit oscillates with the frequency accuracy of the reference oscillation frequency, and the characteristics requirements of the voltage-controlled oscillation circuit can be alleviated.
(18) Here, in the frequency synchronization circuit, a crystal oscillator, a resonator, or the like can be used as the original oscillator for generating a reference frequency. However, using an original oscillator such as a crystal oscillator or a resonator in the sensor interface circuit **1** is difficult for the following reasons.
(19) For example, when used for IoT, the sensor interface circuit **1** is required to reduce power consumption; however, when an original oscillator such as a crystal oscillator or a resonator is used, a current according to an oscillation signal of the original oscillator continues to be passed. Consequently, the current consumption of the sensor interface circuit **1** is increased, and it may be difficult to satisfy the requirement for power consumption reduction.
(20) Each sensor module **100** may generate power by using an RF signal received from the information collection terminal **200**, but the time from power generation to operation is limited due to the restriction of the Radio Law. In the frequency synchronization circuit, when an original oscillator such as a crystal oscillator or a resonator is used, the lock time until the frequency of the reference frequency signal is stabilized is long, and hence it may be difficult to satisfy the restriction of the Radio Law.
(21) An original oscillator such as a crystal oscillator or a resonator tends to be physically larger than a circuit element such as a transistor. Therefore, when an original oscillator such as crystal is used in the frequency synchronization circuit, the sensor interface circuit **1** tends to increase in size, and it may be difficult to satisfy the requirement of the implementation size required for the IoT technology, for example.
(22) An original oscillator such as a crystal oscillator or a resonator is higher in cost than a circuit element such as a transistor. Therefore, when an original oscillator such as a crystal oscillator or a resonator is used in the frequency synchronization circuit, the cost of the sensor interface circuit **1** tends to increase.
(23) Thus, in the first embodiment, in the sensor interface circuit **1**, the frequency synchronization circuit is configured to perform an operation of feedback of a frequency synchronization loop in the voltage domain instead of the frequency domain, and thereby it is attempted to increase the accuracy of the frequency synchronization circuit without using an original oscillator.
(24) Specifically, in the sensor interface circuit **1**, the frequency synchronization circuit generates a reference voltage instead of a reference frequency signal. The reference voltage can be generated by a circuit element such as a resistive element. The frequency synchronization circuit converts the oscillation frequency of an oscillation signal generated by an oscillation operation into impedance. For example, the oscillation frequency of an oscillation signal can be converted into impedance by

using a switched capacitor circuit. The converted impedance is further converted into voltage by using a current according to a detection value of the sensor. The converted voltage is a voltage according to the detection value of the sensor. The operation of feedback of a frequency synchronization loop is performed such that the difference between the voltage according to the detection value of the sensor and the reference voltage approaches zero. Thereby, a frequency synchronization function can be achieved in the voltage domain, and an oscillation frequency less affected by the accuracy of the reference voltage and having high accuracy and high robustness can be obtained. As a result, the accuracy of the frequency synchronization circuit can be increased without using an original oscillator. Therefore, for example, the current consumption of the sensor interface circuit **1** can be reduced, and the requirement of power consumption reduction can be satisfied. The lock time of the frequency synchronization loop can be shortened, and the restriction of the Radio Law can be satisfied. Since a size increase of the sensor interface circuit **1** can be suppressed and the requirement of the implementation size required for the IoT technology can be satisfied, also integration into an integrated circuit is possible without degrading the accuracy of oscillation frequency. Further, the cost of the sensor interface circuit **1** can be made lower than when an original oscillator is used.

(25) More specifically, a sensor interface circuit **1** including a frequency synchronization circuit **10** can be configured as illustrated in FIG. **2**. FIG. **2** is a diagram illustrating a configuration of the sensor module **100** including the sensor interface circuit **1**.

(26) In the sensor module **100**, the sensor interface circuit **1** is connected between the sensor **2**, and the impedance conversion circuit **4** and the antenna **5**. The impedance conversion circuit **4** is electrically connected between the sensor interface circuit **1** and the antenna **5**.

(27) The sensor **2** is a resistive sensor, and includes a variable resistive element R.sub.SENS of which the resistance value changes equivalently according to a detection value of the sensor. One end of the variable resistive element R.sub.SENS is connected to a terminal **1***a* of the sensor interface circuit **1**, and the other end is connected to the ground potential.

(28) The impedance conversion circuit **4** includes an impedance converter **4***a* and an impedance converter **4***b*. The impedance converter **4***a* and the impedance converter **4***b* are connected in parallel between the sensor interface circuit **1** and the antenna **5**. One end of the impedance converter **4***a* is connected to the antenna **5**, and the other end is connected to a terminal **1***b* of the sensor interface circuit **1**. One end of the impedance converter **4***b* is connected to the antenna **5**, and the other end is connected to a terminal **1***c* of the sensor interface circuit **1**.

(29) The sensor interface circuit **1** includes a frequency synchronization circuit **10**, a low-pass filter (LPF) **7**, an RF switch **6**, a power generation circuit **8**, and a voltage control circuit **9**.

(30) The power generation circuit **8** is electrically connected between the terminal **1***c* and the voltage control circuit **9**. The power generation circuit **8** includes an accumulation element such as a capacitive element, and an RF signal is transmitted from the antenna **5** to the power generation circuit **8** via the impedance converter **4***b* and the terminal **1***c*.

(31) The voltage control circuit **9** is electrically connected between the power generation circuit **8** and the frequency synchronization circuit **10**. The voltage control circuit **9**, in a charge period, cuts off power supply to the frequency synchronization circuit **10** to allow charge to be accumulated in the accumulation element, and in a discharge period, uses the charge accumulated in the accumulation element to supply power to the frequency synchronization circuit **10**. Thereby, a power supply potential V.sub.DD can be supplied to an input node **10***a* of the frequency synchronization circuit **10**.

(32) In the frequency synchronization circuit **10**, the input node **10***a* is electrically connected to the voltage control circuit **9**, an input node **10***b* is electrically connected to the sensor **2** via the terminal **1***a*, and an output node **10***c* is electrically connected to the LPF 7. The frequency synchronization circuit **10** performs an oscillation operation according to a detection value of the sensor **2** (for example, a resistance value of the variable resistive element R.sub.SENS). At this time, the

frequency synchronization circuit **10** performs an operation of feedback of a frequency synchronization loop in the voltage domain.

(33) The frequency synchronization circuit **10** generates a reference voltage V.sub.REF. In addition, the frequency synchronization circuit **10** converts the oscillation frequency F.sub.OUT of an oscillation signal generated by the oscillation operation into impedance. The frequency synchronization circuit **10** further converts the converted impedance into a voltage V.sub.SENS by using a current I.sub.SENS according to the resistance value of the variable resistive element R.sub.SENS. The converted voltage V.sub.SENS is a voltage according to the detection value of the sensor. The frequency synchronization circuit **10** performs the operation of feedback of a frequency synchronization loop such that the difference between the voltage V.sub.SENS according to the detection value of the sensor and the reference voltage V.sub.REF approaches 0. Thereby, a frequency synchronization function can be achieved with high accuracy in the voltage domain.

(34) For example, as illustrated in FIG. **2**, the frequency synchronization circuit **10** includes a voltage-controlled oscillation circuit **11**, a frequency division circuit **12**, a frequency/impedance conversion circuit **13**, a reference voltage source **14**, a current source **15**, a voltage difference detection circuit **16**, and a filter **17**. The voltage difference detection circuit **16**, the filter **17**, the voltage-controlled oscillation circuit **11**, the frequency division circuit **12**, and the frequency/impedance conversion circuit **13** are connected in a loop form. This loop-formed connection forms a frequency synchronization loop. In the frequency synchronization circuit **10**, the sensor **2** is connected to the current source **15**, and the current I.sub.SENS passed by the current source **15** changes according to the detection value of the sensor **2**. The frequency synchronization circuit **10** can be referred to as a current change-type frequency synchronization circuit.

(35) The reference voltage source **14** is connected in parallel to the current source **15** and the voltage difference detection circuit **16**. In the reference voltage source **14**, an output node **14***a* is connected to a control node **15***c* of the current source **15** and an input node **16***a* of the voltage difference detection circuit **16**. The reference voltage source **14** generates a reference voltage. For example, the reference voltage source **14** generates a reference voltage V.sub.REF by resistive voltage division as illustrated in FIG. **3**. FIG. **3** is a diagram illustrating a configuration of the frequency synchronization circuit **10**. The reference voltage source **14** includes a plurality of resistive elements **141** and **142**. One end of the resistive element **141** is connected to the ground potential, and the other end is connected to the resistive element **142**. One end of the resistive element **142** is connected to the resistive element **141**, and the other end is connected to a power supply potential V.sub.DD. Assuming that the resistance value of the resistive element **141** is R and the resistance value of the resistive element **142** is R, the reference voltage source **14** can generate a reference voltage V.sub.RE shown in Mathematical Formula 2 below.

$V\text{.sub.REF}=\{R/(R+R)\}\times V\text{.sub.DD}=V\text{.sub.DD}/2 \qquad (2)$

(36) Although FIG. **3** illustrates, as an example, a configuration in which the resistive voltage division ratio is ½, the reference voltage source **14** may be configured with another resistive voltage division ratio value larger than 0 and smaller than 1 according to a required value of the reference voltage V.sub.REF.

(37) The reference voltage source **14** illustrated in FIG. **2** supplies the reference voltage V.sub.REF to the current source **15** and the voltage difference detection circuit **16**.

(38) The current source **15** is electrically connected to the reference voltage source **14**, the voltage difference detection circuit **16**, and the frequency/impedance conversion circuit **13**, and can be electrically connected to the sensor **2**. In the current source **15**, an input node **15***a* is connected to the voltage control circuit **9**, an input node **15***c* is connected to the reference voltage source **14**, an input node **15***d* is connected to the sensor **2**, and an output node **15***b* is connected to an input node **16***b* of the voltage difference detection circuit **16**. The current source **15** generates a current I.sub.SENS according to the detection value of the sensor **2**, and passes the current I.sub.SENS to the input node **16***b* of the voltage difference detection circuit **16**.

(39) For example, the current source **15** can be configured as illustrated in FIG. **3**. The current source **15** includes a transistor **151**, a transistor **152**, and a differential amplification circuit **153**. The transistor **151** is electrically connected between the power supply potential V.sub.DD and the input node **16***b* of the voltage difference detection circuit **16**. The transistor **151** is, for example, a PMOS transistor; the source of the transistor **151** is connected to the power supply potential V.sub.DD, the drain is connected to the input node **16***b* of the voltage difference detection circuit **16**, and the gate is connected to an output node **153***c* of the differential amplification circuit **153**. The transistor **152** can be electrically connected between the power supply potential V.sub.DD and the sensor **2**. The transistor **152** is, for example, a PMOS transistor; the source of the transistor **152** is connected to the power supply potential V.sub.DD, the drain is connected to one end of the variable resistive element R.sub.SENS in the sensor **2**, and the gate is connected to the output node **153***c* of the differential amplification circuit **153** and the gate of the transistor **151**. The differential amplification circuit **153** includes an input node **153***a*, an input node **153***b*, and the output node **153***c*. The input node **153***a* is electrically connected to the reference voltage source **14**, and receives the reference voltage V.sub.REF from the reference voltage source **14**. The input node **153***b* is electrically connected to a node **154** between the transistor **152** and the sensor **2**. The output node **153***c* is connected in common to the gate of the transistor **151** and the gate of the transistor **152**.
(40) That is, the transistor **151** and the transistor **152** constitute a current mirror circuit via the differential amplification circuit **153**. Using a feedback loop of the differential amplification circuit **153**.fwdarw.the transistor **152**.fwdarw.the node **154**.fwdarw.the differential amplification circuit **153**, the differential amplification circuit **153** controls the gate voltage of the transistor **151** and the gate voltage of the transistor **152** such that the potential of the node **154** becomes equal to the reference voltage V.sub.REF. Thus, the current I.sub.SENS′ flowing through the variable resistive element R.sub.SENS is expressed by Mathematical Formula 3 below.
I.sub.SENS′=*V*.sub.REF/*R*.sub.SENS (3)
(41) Assuming that the mirror ratio of the current mirror circuit is 1, the current I.sub.SENS flowing from the current source **15** to the input node **16***b* of the voltage difference detection circuit **16** is expressed by Mathematical Formula 4 below.
I.sub.SENS=*I*.sub.SENS=*V*.sub.REF/*R*.sub.SENS (4)
(42) As shown in Mathematical Formula 4, the current I.sub.SENS of the current source **15** changes according to the change in the resistance value R.sub.SENS of the sensor **2**. The current I.sub.SENS indicates a change in the resistance value R.sub.SENS of the sensor **2**.
(43) The voltage-controlled oscillation circuit **11** illustrated in FIG. **2** is electrically connected between the voltage difference detection circuit **16** and the output node **10***c*, and is electrically connected between the filter **17** and the frequency division circuit **12**. In the voltage-controlled oscillation circuit **11**, an input node **11***a* is electrically connected to an output node **16***c* of the voltage difference detection circuit **16** via the filter **17**, and an output node **11***b* is electrically connected to the output node **10***c* via the frequency division circuit **12**. The voltage-controlled oscillation circuit **11** performs an oscillation operation according to a control voltage V.sub.CTRL received from the voltage difference detection circuit **16** via the filter **17**, and generates an oscillation signal having a frequency F.sub.SENS according to the control voltage V.sub.CTRL.
(44) For example, as illustrated in FIG. **3**, the voltage-controlled oscillation circuit **11** can be configured with a relaxation oscillation circuit. The voltage-controlled oscillation circuit **11** includes an inverter chain **111**, a variable resistive element **112**, and a capacitive element **113**. The inverter chain **111** includes a plurality of stages of inverters Inv1 to Inv3 connected in a ring form. Each inverter Inv is configured by, for example, connecting an NMOS transistor and a PMOS transistor in an inverter connection manner. The number of stages of inverters Inv is an odd number, for example, three. An output node of the first-stage inverter Inv1 is electrically connected to an input node of the next-stage inverter Inv2. An output node of the last-stage inverter Inv3 is electrically connected to the output node **11***b* of the voltage-controlled oscillation circuit **11** and an

input node of the first-stage inverter Inv1. The variable resistive element **112** is electrically connected in series to the plurality of stages of inverters Inv1 to Inv3 in the inverter chain **111**. FIG. **3** illustrates, as an example, a configuration in which the variable resistive element **112** is electrically connected between the output node of the inverter Inv2 and the input node of the inverter Inv3. The capacitive element **113** is connected in parallel to an inverter Inv and the variable resistive element **112** in the inverter chain **111**. FIG. **3** illustrates, as an example, a configuration in which the capacitive element **113** is connected in parallel to a series connection of the second-stage inverter Inv2 and the variable resistive element **112**.
(45) In the voltage-controlled oscillation circuit **11**, the variable resistive element **112** receives the control voltage V.sub.CTRL at a control node. The resistance value R.sub.VCO of the variable resistive element **112** changes according to the control voltage V.sub.CTRL. Assuming that the capacitance value of the capacitive element **113** is C.sub.VCO, the time constant R.sub.VCO×C.sub.CVO of the variable resistive element **112** and the capacitive element **113** changes with the change in the resistance value R.sub.VCO of the variable resistive element **112**. The oscillation frequency F.sub.SENS of the voltage-controlled oscillation circuit **11** is determined according to the time constant R.sub.VCO×C.sub.VCO. That is, in the voltage-controlled oscillation circuit **11**, the time constant R.sub.VCO×C.sub.VCO changes according to the control voltage V.sub.CTRL, and an oscillation signal having a frequency F.sub.SENS according to the time constant R.sub.VCO×C.sub.VCO after change is generated. The variable resistive element **112** may be configured with, for example, an NMOS transistor in which the drain is connected to the output node of the inverter Inv2, the source is connected to the input node of the inverter Inv3, and the control voltage V.sub.CTRL is applied the gate.
(46) The voltage-controlled oscillation circuit **11** illustrated in FIG. **2** supplies an oscillation signal having the frequency F.sub.SENS to the frequency division circuit **12**.
(47) The frequency division circuit **12** is electrically connected between the voltage-controlled oscillation circuit **11** and the output node **10***c*. In the frequency division circuit **12**, an input node **12***a* is electrically connected to the output node **11***b* of the voltage-controlled oscillation circuit **11**, and an output node **12***b* is electrically connected to the output node **10***c*. The frequency division circuit **12** performs frequency division on an oscillation signal received at the input node **12***a* and generates an oscillation signal having a frequency F.sub.OUT, and supplies the oscillation signal to the LPF 7 and the frequency/impedance conversion circuit **13**. The frequency F.sub.OUT may be ½ of the frequency F.sub.SENS. At this time, the frequency division circuit **12** can adjust the duty ratio of the oscillation signal to, for example, around 50%.
(48) For example, as illustrated in FIG. **3**, the frequency division circuit **12** is configured to perform frequency division by two. The frequency division circuit **12** includes a flip-flop **121** and an inverter **122**. In the flip-flop **121**, a data input node D is connected to an output node of the inverter **122**, a clock node CK is connected to the output node **11***b* of the voltage-controlled oscillation circuit **11**, and a data output node Q is connected to an input node of the inverter **122**, the LPF 7, and the frequency/impedance conversion circuit **13**. The flip-flop **121** holds a signal in which an output signal of its own is logically inverted in synchronization with the rising edge of the waveform of the oscillation signal, and toggles output signals of its own. Thereby, the frequency division circuit **12** performs frequency division by two on the oscillation signal received at the input node **12***a*, and generate an oscillation signal having a frequency F.sub.OUT (=F.sub.SENS/2). Further, the frequency division circuit **12** can adjust the duty ratio of the oscillation signal to around 50% by toggling oscillation signals outputted with the period of the received oscillation signal.
(49) The frequency division circuit **12** illustrated in FIG. **2** outputs an oscillation signal having the frequency F.sub.OUT to the LPF 7, and feeds back the oscillation signal to the frequency/impedance conversion circuit **13**.
(50) The frequency/impedance conversion circuit **13** is electrically connected between the voltage-controlled oscillation circuit **11** and the input node **16***b* of the voltage difference detection circuit

16, and is electrically connected between the frequency division circuit **12** and the input node **16***b* of the voltage difference detection circuit **16**. The frequency/impedance conversion circuit **13** is provided on a feedback line from the output node **12***b* of the frequency division circuit **12** to the input node **16***b* of the voltage difference detection circuit **16**. In the frequency/impedance conversion circuit **13**, an input node **13***a* is connected to the frequency division circuit **12**, and an output node **13***b* is connected to the input node **16***b* of the voltage difference detection circuit **16**. The frequency/impedance conversion circuit **13** receives an oscillation signal from the frequency division circuit **12**, and converts the frequency F.sub.OUT of the oscillation signal into impedance.
(51) For example, as illustrated in FIG. **3**, the frequency/impedance conversion circuit **13** can be configured with a switched capacitor circuit. The switched capacitor circuit is a circuit that limits current or voltage like a resistor by combining a switch and a capacitive element. The frequency/impedance conversion circuit **13** can set the circuit impedance to a value according to the frequency of the oscillation signal by charging and discharging the capacitive element according to the oscillation signal.
(52) The frequency/impedance conversion circuit **13** includes a capacitive element **131**, a capacitive element **132**, a switch **133**, a switch **134**, and an inverter **135**. One end of the capacitive element **131** is connected to the ground potential, and the other end is connected to a node **136** between the switch **133** and the switch **134**. One end of the capacitive element **132** is connected to the ground potential, and the other end is connected to the input node **16***b* of the voltage difference detection circuit **16**. One end of the switch **133** is connected to the input node **16***b* of the voltage difference detection circuit **16**, the other end is connected to the node **136**, and a control end is connected to the voltage-controlled oscillation circuit **11** via the frequency division circuit **12**. One end of the switch **134** is connected to the node **136**, the other end is connected to the ground potential, and a control end is connected to the inverter **135**. In the inverter **135**, an input node is connected to the voltage-controlled oscillation circuit **11** via the frequency division circuit **12**, and an output node is connected to the switch **134**.
(53) In the frequency/impedance conversion circuit **13**, the switch **133** and the switch **134** are complementarily turned on and off according to the level of the oscillation signal. Thereby, the capacitive element **131** is charged and discharged. When the oscillation signal is at an H level, the switch **133** is maintained in the off state and the switch **134** is maintained in the on state, the charge of the capacitive element **131** is discharged to the ground potential, and the capacitive element **131** is discharged. When the oscillation signal is at an L level, the switch **133** is maintained in the on state and the switch **134** is maintained in the off state, a charge according to the current I.sub.SENS is accumulated in the capacitive element **131**, and the capacitive element **131** is charged. At this time, the capacitive element **132** maintains a state where a charge according to the current I.sub.SENS is accumulated regardless of the level of the oscillation signal.
(54) That is, the frequency/impedance conversion circuit **13** can generate an impedance equivalently corresponding to the frequency F.sub.OUT by periodically repeating charge and discharge on the capacitive element **131** by means of an oscillation signal having the frequency F.sub.OUT. The output voltage of the frequency/impedance conversion circuit **13** appears as the voltage V.sub.SENS of the input node **16***b* of the voltage difference detection circuit **16**. The voltage V.sub.SENS changes in a time constant manner during charge of the capacitive element **131**, but converges to a stable point while being averaged by the capacitive element **132** that maintains accumulation of a charge according to the current I.sub.SENS.
(55) For example, as illustrated in FIG. **4**, when the current I.sub.SENS=I.sub.1, then V.sub.SENS=V.sub.1, which is a voltage at the time of convergence to a stable point. When the current I.sub.SENS=I.sub.2, then V.sub.SENS=V.sub.2, which is a voltage at the time of convergence to a stable point. When the current I.sub.SENS=10, then V.sub.SENS=V.sub.10, which is a voltage at the time of convergence to a stable point. FIG. **4** is a diagram illustrating an operation of the frequency/impedance conversion circuit **13**, where the vertical axis represents the

magnitude of voltage and the horizontal axis represents time. In the frequency/impedance conversion circuit **13**, it can be seen that when the flowing-in current I.sub.SENS increases, the voltage V.sub.SENS at the time of convergence to a stable point rises almost in proportion thereto. That is, assuming that the capacitance value of the capacitive element **132** is C.sub.AVE and the capacitance value of the capacitive element **131** is C.sub.SC, the voltage V.sub.SENS of the input node **16***b* of the voltage difference detection circuit **16** at the time of convergence to a stable point is expressed by Mathematical Formula 5 below.

V.sub.SENS=*I*.sub.SENS/(*F*.sub.OUT.*Math.C*.sub.SC) (5)

(56) As shown in Mathematical Formula 5, in a state of convergence to a stable point, the frequency F.sub.OUT of the oscillation signal is converted into an impedance 1/(F.sub.OUT.Math.C.sub.SC) by the frequency/impedance conversion circuit **13**. Equivalently, one end of the impedance 1/(F.sub.OUT.Math.C.sub.SC) is connected to the input node **16***b* of the voltage difference detection circuit **16**, and the other end is connected to the ground potential. Therefore, at the input node **16***b* of the voltage difference detection circuit **16**, the current I.sub.SENS from the current source **15** flows into the equivalent impedance 1/(F.sub.OUT.Math.C.sub.SC), and thereby the current I.sub.SENS is converted into the voltage V.sub.SENS by the equivalent impedance 1/(F.sub.OUT.Math.C.sub.SC). The Voltage V.sub.SENS Includes the Current I.sub.SENS, and corresponds to the detection value of the sensor **2**. Further, the voltage V.sub.SENS includes the frequency F.sub.CUT, and corresponds to the oscillation frequency F.sub.SENS of the voltage-controlled oscillation circuit **11**.

(57) The reference voltage source **14**, the current source **15**, and the frequency/impedance conversion circuit **13** are connected to input terminals of the voltage difference detection circuit **16** illustrated in FIG. **2**, and the filter **17** is electrically connected to an output terminal of the voltage difference detection circuit **16**. In the voltage difference detection circuit **16**, the input node **16***a* is connected to the reference voltage source **14**, the input node **16***b* is connected to the current source **15** and the frequency/impedance conversion circuit **13**, and the output node **16***c* is connected to the filter **17**. The voltage difference detection circuit **16** receives the reference voltage V.sub.REF at the input node **16***a*, and a voltage V.sub.SENS is generated at the input node **16***b* by the current source **15** and the frequency/impedance conversion circuit **13**. According to the difference between the reference voltage V.sub.REF and the voltage V.sub.SENS, the voltage difference detection circuit **16** generates a control voltage V.sub.CTRL′ for making control such that the difference becomes smaller.

(58) For example, as illustrated in FIG. **3**, the voltage difference detection circuit **16** includes a differential amplification circuit **161**. In the differential amplification circuit **161**, a non-inverting input terminal (+) is connected to the reference voltage source **14**, an inverting input terminal (−) is connected to the current source **15** and the frequency/impedance conversion circuit **13**, and an output terminal is connected to the filter **17**. The differential amplification circuit **161** amplifies the difference between the reference voltage V.sub.REF and the voltage V.sub.SENS to generate a control voltage V.sub.CTRL′.

(59) The filter **17** illustrated in FIG. **2** is electrically connected between the voltage difference detection circuit **16** and the voltage-controlled oscillation circuit **11**. In the filter **17**, an input node **17***a* is connected to the voltage difference detection circuit **16**, and an output node **17***b* is connected to the voltage-controlled oscillation circuit **11**. The filter **17** receives the control voltage V.sub.CTRL′ from the voltage difference detection circuit **16**, and performs filtering processing on the control voltage V.sub.CTRL′. The filter **17** supplies the control voltage V.sub.CTRL after filtering processing to the voltage-controlled oscillation circuit **11**.

(60) For example, as illustrated in FIG. **3**, the filter **17** can be configured with a low-pass filter. The filter **17** includes a resistive element **171** and a capacitive element **172**. One end of the resistive element **171** is connected to on output terminal of the differential amplification circuit **161**, and the other end is connected to one end of the capacitive element **172** and the voltage-controlled

oscillation circuit **11**. The other end of the capacitive element **172** is connected to the ground potential. With this configuration, the filter **17** can subject the control voltage V.sub.CTRL′ to Low-Pass Filtering Processing for smoothing, and supply the smoothed control voltage V.sub.CTRL to the voltage-controlled oscillation circuit **11**.
(61) In the frequency synchronization circuit **10**, using a frequency synchronization loop of the voltage difference detection circuit **16**.fwdarw.the filter **17**.fwdarw.the voltage-controlled oscillation circuit **11**.fwdarw.the frequency division circuit **12**.fwdarw.the frequency/impedance conversion circuit **13**.fwdarw.the voltage difference detection circuit **16**, the voltage difference detection circuit **16** performs feedback control on the control voltage V.sub.CTRL such that the voltage V.sub.SENS becomes equal to the reference voltage V.sub.REF. That is, when the feedback control works normally, Mathematical Formula 6 below holds.
V.sub.REF=*V*.sub.SENS (6)
(62) When Mathematical Formula 6 is substituted into Mathematical Formula 5, Mathematical Formula 7 below is obtained.
V.sub.REF=*I*.sub.SENS/(*F*.sub.OUT.*Math.C*.sub.SC) (7)
(63) When Mathematical Formula 7 is solved for the frequency F.sub.OUT, Mathematical Formula 8 below is obtained.
F.sub.OUT=1/{(*V*.sub.REF/*I*.sub.SENS).Math.*C*.sub.SC} (8)
(64) When Mathematical Formula 4 is substituted into Mathematical Formula 8, Mathematical Formula 9 below is obtained.
F.sub.OUT=1/{*R*.sub.SENS.*Math.C*.sub.SC} (9)
(65) As shown in Mathematical Formula 9, it can be seen that, in the frequency synchronization circuit **10**, an oscillation frequency F.sub.OUT according to an equivalent resistance value R.sub.SENS of the sensor **2** is obtained without depending on the reference voltage V.sub.REF. Thereby, a frequency synchronization function can be achieved in the voltage domain, and an oscillation frequency F.sub.OUT less affected by the accuracy of the reference voltage V.sub.REF can be obtained. That is, an oscillation frequency F.sub.OUT with high accuracy and high robustness can be obtained.
(66) Further, as shown in Mathematical Formula 9, in the frequency synchronization circuit **10**, an oscillation frequency F.sub.OUT insensitive to the power supply potential V.sub.DD can be obtained. Thereby, resistance to fluctuation of the power supply potential V.sub.DD can be enhanced, and characteristics requirements on the voltage control circuit **9** can be alleviated. As a result, when the sensor interface circuit **1** is implemented with a semiconductor chip, the chip cost can be reduced.
(67) Further, as shown in Mathematical Formula 9, in the frequency synchronization circuit **10**, an oscillation frequency F.sub.OUT insensitive to the variation in characteristics of the transistor can be obtained. Thereby, characteristics requirements on the transistor can be alleviated. As a result, when the sensor interface circuit **1** is implemented with a semiconductor chip, the chip cost can be reduced.
(68) Further, as shown in Mathematical Formula 9, in the frequency synchronization circuit **10**, an oscillation frequency F.sub.OUT insensitive to temperature can be obtained by obtaining an oscillation frequency F.sub.OUT depending on a resistance value and a capacitance value with small temperature characteristics. Thereby, resistance to temperature fluctuation can be improved with inexpensive components without using a crystal oscillator or a MEMS resonator. As a result, when the sensor interface circuit **1** is implemented with a semiconductor chip, the chip cost can be reduced.
(69) Next, an operation of the frequency synchronization circuit **10** is described using FIG. **5**. FIG. **5** is a waveform diagram illustrating an operation of the frequency synchronization circuit **10**.
(70) At timing t1, when the frequency synchronization circuit **10** is activated, the current source **15** starts to pass a current I.sub.SENS=I.sub.1 according to a resistance value R.sub.SENS=R.sub.1 of

the sensor **2**. In accordance with this, in the frequency/impedance conversion circuit **13**, a charge according to the current I.sub.SENS=I.sub.1 is accumulated in the capacitive element **132**, and the voltage V.sub.SENS rises gradually. At this time, the oscillation signal is at the L level, the switch **133** is maintained in the off state, the switch **134** is maintained in the on state, the capacitive element **131** is in the discharge state, and the voltage V.sub.SC thereof is substantially equal to the ground potential. Further, since the reference voltage V.sub.REF is maintained fixed, the voltage difference detection circuit **16** gradually increases the control voltage V.sub.CTRL as the voltage V.sub.SENS rises gradually.
(71) At timing t2, the voltage-controlled oscillation circuit **11** starts an oscillation operation, and the level of the oscillation signal becomes the H level. In accordance with this, the switch **133** is turned on, and the switch **134** is turned off. The voltage V.sub.SENS is lowered momentarily due to the fact that one end of the capacitive element **131** is connected to the input node **16***b*. After that, charge is redistributed between the capacitive elements **131** and **132**, and a charge according to the current I.sub.SENS=I.sub.1 is injected into each of the capacitive elements **131** and **132**; accordingly, the voltage V.sub.SC and the voltage V.sub.SENS rise gradually.
(72) At timing t3, the level of the oscillation signal becomes the L level. In accordance with this, the switch **133** is turned off, and the switch **134** is turned on. The capacitive element **131** is discharged, and its voltage V.sub.SC is lowered to the ground potential. At this time, since the capacitive element **132** holds charge, the voltage V.sub.SENS continues to rise gradually in accordance with a charge according to the current I.sub.SENS=I.sub.1 being injected into the capacitive element **132**.
(73) At timing t4, the level of the oscillation signal becomes the H level. In accordance with this, the switch **133** is turned on, and the switch **134** is turned off. The voltage V.sub.SENS is lowered momentarily due to the fact that one end of the capacitive element **131** is connected to the input node **16***b*. After that, charge is redistributed between the capacitive elements **131** and **132**, and a charge according to the current I.sub.SENS=I.sub.1 is injected into each of the capacitive elements **131** and **132**; accordingly, the voltage V.sub.SC and the voltage V.sub.SENS rise gradually.
(74) At timing t5, the level of the oscillation signal becomes the L level. In accordance with this, the switch **133** is turned off, and the switch **134** is turned on. The capacitive element **131** is discharged, and its voltage V.sub.SC is lowered to the ground potential. At this time, since the capacitive element **132** holds charge, the voltage V.sub.SENS continues to rise gradually in accordance with a charge according to the current I.sub.SENS=I.sub.1 being injected into the capacitive element **132**.
(75) At timings t6 to t14, similar operations to those at timings t4 and t5 and similar operations to those at timings t5 and t6 are alternately repeated; and the voltage V.sub.SENS approaches the reference voltage V.sub.REF on a time average basis while fluctuating in a sawtooth-like waveform, and the control voltage V.sub.CTRL gradually approaches the value V.sub.1. Accordingly, the frequency F.sub.OUT of the oscillation signal approaches a value F.sub.1 corresponding to the resistance value R.sub.SENS=R.sub.1 of the sensor **2**.
(76) When at timing t14 the voltage V.sub.SENS becomes equal to the reference voltage V.sub.REF on a time average basis, then from timing t14 to timing t18, the control voltage V.sub.CTRL is stabilized at the value V.sub.1, the frequency F.sub.OUT of the oscillation signal is stabilized at a value F.sub.1 corresponding to the resistance value R.sub.SENS=R.sub.1 of the sensor **2**, and the frequency synchronization loop is set in a locked state. Thereby, the frequency synchronization circuit **10** stably outputs a frequency F.sub.OUT=F.sub.1 corresponding to the resistance value R.sub.SENS=R.sub.1 of the sensor **2**. The period TP1 is a period corresponding to the frequency F.sub.1.
(77) At timing t18, the resistance value R.sub.SENS of the sensor **2** changes to R.sub.2 (>R.sub.1) due to a state change or the like of the detection target of the sensor **2**, and the current changes to the current I.sub.SENS=1.sub.2 (<I.sub.1). In accordance with this, at timings t18 to t24, similar

operations to those at timings t4 and t5 and similar operations to those at timings t5 and t6 are alternately repeated; and the voltage V.sub.SENS approaches the reference voltage V.sub.REF on a time average basis while fluctuating in a sawtooth-like waveform, and the control voltage V.sub.CTRL gradually approaches the value V.sub.2 (>V.sub.1). Accordingly, the frequency F.sub.OUT of the oscillation signal approaches a value F.sub.2 (<F.sub.1) corresponding to the resistance value R.sub.SENS=R.sub.2 of the sensor **2**.
(78) When at timing t24 the voltage V.sub.SENS becomes substantially equal to the reference voltage V.sub.REF on a time average basis, then on or after timing t24, the control voltage V.sub.CTRL is stabilized at the value V.sub.2, the frequency F.sub.OUT of the oscillation signal is stabilized at a value F.sub.2 corresponding to the resistance value R.sub.SENS=R.sub.2 of the sensor **2**, and the frequency synchronization loop is set in a locked state again. Thereby, the frequency synchronization circuit **10** stably outputs a frequency F.sub.UT=F.sub.2 corresponding to the resistance value R.sub.SENS=R.sub.2 of the sensor **2**. The period TP2 is a period corresponding to the frequency F.sub.2.
(79) As above, in the first embodiment, in the sensor interface circuit **1**, the frequency synchronization circuit **10** is configured to perform an operation of feedback of a frequency synchronization loop in the voltage domain instead of the frequency domain. Thereby, an oscillation frequency not depending on the accuracy of the reference voltage can be obtained, and the accuracy of the operation of feedback of a frequency synchronization loop can be easily increased in the voltage domain. Therefore, the accuracy of the frequency synchronization circuit **10** can be increased without using an original oscillator.
(80) In the frequency synchronization circuit **10**, when the duty ratio of the oscillation signal outputted from the voltage-controlled oscillation circuit **11** is close to 50%, the frequency division circuit **12** may be omitted. Further, when the control voltage outputted from the voltage difference detection circuit **16** is smooth, the filter **17** may be omitted.
Second Embodiment
(81) Next, a sensor interface circuit **1***j* according to a second embodiment is described. In the following, portions different from those of the first embodiment are mainly described.
(82) As a frequency synchronization circuit that performs an operation of feedback of a frequency synchronization loop in the voltage domain, in the first embodiment a current change-type frequency synchronization circuit is given as an example, whereas in the second embodiment a voltage change-type frequency synchronization circuit is given as an example. With respect to the change in the detection value of the sensor, the current of a current source changes in the current change-type frequency synchronization circuit, whereas the voltage amplified and outputted by an amplification circuit is changed in the voltage change-type frequency synchronization circuit.
(83) Specifically, a sensor module **100***j* including the sensor interface circuit **1***j* can be configured as illustrated in FIG. **6**. FIG. **6** is a diagram illustrating a configuration of the sensor module **100***j* including the sensor interface circuit **1***j* according to the second embodiment.
(84) The sensor interface circuit **1***j* includes a frequency synchronization circuit **10***j* instead of the frequency synchronization circuit **10** (see FIG. **2**). The frequency synchronization circuit **10***j* includes a current source **15***j* instead of the current source **15** (see FIG. **2**), and further includes an amplification circuit **18***j* and a resistive element **19***j*.
(85) The amplification circuit **18***j* is electrically connected between the reference voltage source **14** and the voltage difference detection circuit **16**, and can be electrically connected to the sensor **2**. In the amplification circuit **18***j*, an input node **18***a* is connected to the reference voltage source **14**, an input node **18***b* is connected to a node **192** between the resistive element **19***j* and the variable resistive element R.sub.SENS, and an output node **18***c* is connected to the voltage difference detection circuit **16**.
(86) The reference voltage source **14** is connected to the current source **15***j* and the amplification circuit **18***j*. In the reference voltage source **14**, an output node **14***a* is connected to a control node

15*c* of the current source **15***j* and the input node **18***a* of the amplification circuit **18***j*. The reference voltage source **14** generates a reference voltage V.sub.REF′. The amplification circuit **18***j* receives the reference voltage V.sub.REF′ at the input node **18***a*.
(87) One end of the variable resistive element R.sub.SENS of the sensor **2** is connected to the voltage control circuit **9** via a terminal **1***d* of the sensor interface circuit **1***j*, and the other end is connected to the amplification circuit **18***j* and the resistive element **19***j* via a terminal **1***e* of the sensor interface circuit **1***j*. A power supply potential V.sub.DD is supplied to one end of the variable resistive element R.sub.SENS, and the other end is connected to one end of the resistive element **19***j*. The other end of the resistive element **19***j* is connected to the ground potential. The resistive element **19***j* is a reference resistive element, and has a resistance value serving as a reference for change in the resistance value of the variable resistive element R.sub.SENS. Thereby, a voltage V.sub.SENS′ obtained by the power supply potential V.sub.DD being subjected to resistive voltage division according to the resistance value of the variable resistive element R.sub.SENS is applied to the input node **18***b* of the amplification circuit **18***j*. The voltage V.sub.SENS′ changes according to the change in the resistance value R.sub.SENS of the sensor **2**. The voltage V.sub.SENS′ indicates a change in the resistance value R.sub.SENS of the sensor **2**.
(88) The amplification circuit **18***j* amplifies the difference between the reference voltage V.sub.REF′ and the voltage V.sub.SENS′, and outputs a voltage V.sub.SENS according to the amplified difference. The voltage V.sub.SENS corresponds to a detection value of the sensor **2** (for example, a resistance value of the variable resistive element R.sub.SENS) That is, the voltage difference detection circuit **16** is different from that of the first embodiment in that the voltage V.sub.SENS corresponding to the detection value of the sensor **2** is received at the input node **16***a* instead of the input node **16***b* (see FIG. **3**) and the reference voltage V.sub.REF is received at the input node **16***b* instead of the input node **16***a* (see FIG. **3**).
(89) For example, as illustrated in FIG. **7**, the amplification circuit **18***j* is configured with an instrumentation amplification circuit. The amplification circuit **18***j* includes a plurality of drivers **181** to **183** and a plurality of resistive elements **184** to **191**. In the driver **181**, a first input node is connected to the reference voltage source **14**, a second input node is connected to one end of the resistive element **184** and one end of the resistive element **186**, and an output node is connected to one end of the resistive element **188** and the other end of the resistive element **186**. In the driver **182**, a first input node is connected to the node **192**, a second input node is connected to one end of the resistive element **185** and one end of the resistive element **187**, and an output node is connected to one end of the resistive element **191** and the other end of the resistive element **187**. In the driver **183**, a first input node is connected to the other end of the resistive element **188** and one end of the resistive element **189**, a second input node is connected to one end of the resistive element **190** and the other end of the resistive element **191**, and an output node is connected to the other end of the resistive element **189**. The other end of the resistive element **190** is connected to the other end of the resistive element **184** and the other end of the resistive element **185**. The driving forces of the plurality of drivers **181** to **183** may be equal to each other. The resistance values of the plurality of resistive elements **184** to **191** may be equal to each other.
(90) With this configuration, the amplification circuit **18***j* can amplify the difference of the voltage V.sub.SENS′ from the reference voltage V.sub.REF′ to a larger value to generate a voltage V.sub.SENS. Assuming that the resistance value of the resistive element **19***j* is R.sub.REF and the resistance value of the sensor **2** (the variable resistive element R.sub.SENS) is R.sub.SENS, the voltage V.sub.SENS′ is expressed by Mathematical Formula 10 below.
$$V\text{.sub.SENS}'=V\text{.sub.DD.}Math.\{R\text{.sub.REF}/(R\text{.sub.REF}+R\text{.sub.SENS})\} \qquad (10)$$
(91) Further, assuming that the amplification factor of the amplification circuit **18***j* is A, the voltage V.sub.SENS after amplification is expressed by Mathematical Formula 11 below.
$$V\text{.sub.SENS}=(V\text{.sub.SENS}'-V\text{.sub.REF}')\text{.Math.}A+V\text{.sub.REF}' \qquad (11)$$
(92) Since V.sub.REF′=V.sub.DD/2, when Mathematical Formula 10 is substituted into

Mathematical Formula 11, Mathematical Formula 12 below is obtained.

$$V_{SENS}=(V_{DD}\cdot\{R_{REF}/(R_{REF}+R_{SENS})\}-V_{DD}/2)\cdot A+V_{DD}/2 \qquad (12)$$

(93) As shown in Mathematical Formulae 11 and 12, the amplification circuit **18**_j_ can amplify the difference of the voltage V.sub.SENS′ from the reference voltage V.sub.REF′ to a larger value to generate a voltage V.sub.SENS.

(94) The current source **15**_j_ illustrated in FIG. **6** is not connected to the sensor **2**, and passes a reference current I.sub.REF according to the reference voltage V.sub.REF′ to the input node **16**_b_ of the voltage difference detection circuit **16**.

(95) For example, as illustrated in FIG. **7**, the current source **15**_j_ further includes a resistive element **155**_j_. The resistive element **155**_j_ has a fixed resistance value. The present embodiment is similar to the first embodiment in that, using a feedback loop of the differential amplification circuit **153**.fwdarw.the transistor **152**.fwdarw.the node **154**.fwdarw.the differential amplification circuit **153**, the differential amplification circuit **153** controls the gate voltage of the transistor **151** and the gate voltage of the transistor **152** such that the potential of the node **154** becomes equal to the reference voltage V.sub.REF′. Thereby, the current source **15**_j_ can pass a reference current I.sub.REF according to the reference voltage V.sub.REF′ to the input node **16**_b_ of the voltage difference detection circuit **16**. Since V.sub.REF′=V.sub.DD/2, assuming that the resistance value of the resistive element **155**_j_ is R, Mathematical Formula 13 below holds.

$$I_{REF}=(V_{DD}/2)/R \qquad (13)$$

(96) The output voltage of the frequency/impedance conversion circuit **13** appears as the voltage V.sub.REF of the input node **16**_b_ of the voltage difference detection circuit **16**. The voltage V.sub.REF changes in a time constant manner during charge of the capacitive element **131**, but converges to a stable point while being averaged by the capacitive element **132** that maintains accumulation of a charge according to the current I.sub.REF. The voltage V.sub.SENS of the input node **16**_b_ of the voltage difference detection circuit **16** at the time of convergence to a stable point is expressed by Mathematical Formula 14 below.

$$V_{REF}=I_{REF}/(F_{OUT}\cdot C_{SC}) \qquad (14)$$

(97) As shown in Mathematical Formula 14, in a state of convergence to a stable point, the frequency F.sub.OUT of the oscillation signal is converted into an impedance “1/(F.sub.OUT.Math.C.sub.SC)” by the frequency/impedance conversion circuit **13**. Equivalently, one end of the impedance “1/(F.sub.OUT.Math.C.sub.SC)” is connected to the input node **16**_b_ of the voltage difference detection circuit **16**, and the other end is connected to the ground potential. Therefore, at the input node **16**_b_ of the voltage difference detection circuit **16**, the current I.sub.REF from the current source **15**_j_ flows into the equivalent impedance “1/(F.sub.OUT.Math.C.sub.SC)”, and thereby the current I.sub.REF is converted into the voltage V.sub.REF by the equivalent impedance “1/(F.sub.OUT.Math.C.sub.SC)”. The present embodiment is similar to the first embodiment in that the voltage V.sub.REF includes the frequency F.sub.OUT and corresponds to the oscillation frequency F.sub.SENS of the voltage-controlled oscillation circuit **11**, but is different from the first embodiment in that the voltage V.sub.REF does not correspond to the detection value of the sensor **2**. Instead, the voltage V.sub.SENS shown in Mathematical Formula 12 includes the resistance value R.sub.SENS, and corresponds to the detection value of the sensor **2**.

(98) The present embodiment is similar to the first embodiment in that, in the frequency synchronization circuit **10**_j_, using a frequency synchronization loop of the voltage difference detection circuit **16**.fwdarw.the filter **17**.fwdarw.the voltage-controlled oscillation circuit **11**.fwdarw.the frequency division circuit **12**.fwdarw.the frequency/impedance conversion circuit **13**.fwdarw.the voltage difference detection circuit **16**, the voltage difference detection circuit **16** performs feedback control on the control voltage V.sub.CTRL such that the voltage V.sub.SENS becomes equal to the reference voltage V.sub.REF. That is, when the feedback control works

normally, Mathematical Formula 6 holds. When Mathematical Formula 6 is substituted into Mathematical Formula 14, Mathematical Formula 15 below is obtained.

$V_{SENS}=I_{REF}/(F_{OUT}\cdot C_{SC})$ (15)

(99) When Mathematical Formulae 12 and 13 are substituted into Mathematical Formula 15, Mathematical Formula 16 below is obtained.

$(V_{DD}\cdot\{R_{REF}/(R_{REF}+R_{SENS})\}-V_{DD}/2)\cdot A+V_{DD}/2=\{(V_{DD}/2)/R\}\cdot\{1/(F_{OUT}\cdot C_{SC})\}$ (16)

(100) When $\{R_{REF}/(R_{REF}+R_{SENS})\}=RS$ is set in Mathematical Formula 16, Mathematical Formula 16 is rewritten as Mathematical Formula 17 below. RS is a parameter indicating a change in the resistance value R_{SENS} of the sensor **2**.

$(V_{DD}\cdot RS-V_{DD})/2)\cdot A+V_{DD})/2=\{(V_{DD}/2)/R\}\cdot\{1/(F_{OUT}\cdot C_{SC})\}$ (17)

(101) When Mathematical Formula 17 is solved for the frequency F_{OUT}, Mathematical Formula 18 below is obtained.

$F_{OUT}=[\{(V_{DD}/2)/R\}\cdot\{1/(C_{SC})\}]/[(V_{DD}\cdot RS-V_{DD}/2)\cdot A+V_{DD}/2]=\{1/(C_{SC}\cdot R)\}\cdot[1/\{2\cdot A\cdot RS+(1-A)\}]$ (18)

(102) As shown in Mathematical Formula 18, in the sensor interface circuit **1***j*, the oscillation frequency can be designed by the first term $\{1/(C_{SC}\cdot R)\}$ on the right side, and the sensitivity (amplification factor A) to the detection value of the sensor **2** can be designed by the second term $[1/\{2\cdot A\cdot RS+(1-A)\}]$. That is, an oscillation frequency F_{OUT} with higher accuracy and higher robustness can be obtained than in the first embodiment.

(103) Further, as illustrated in FIG. **8**, the operation of the frequency synchronization circuit **10***j* is different from that of the first embodiment in the following point. FIG. **8** is a waveform diagram illustrating an operation of the frequency synchronization circuit **10***j*.

(104) In the waveform diagram of FIG. **8**, in contrast to the waveform diagram of FIG. **5**, what changes in a sawtooth-like form is replaced from the voltage V_{SENS} to the voltage V_{REF}, and what is relatively fixed is replaced from the voltage V_{REF} to the voltage V_{SENS}. The voltage V_{SENS} is almost fixed during a period when the resistance value R_{SENS} of the sensor **2** is fixed, but when the resistance value R_{SENS} of the sensor **2** changes, changes accordingly.

(105) At timing t31, when the frequency synchronization circuit **10***j* is activated, the current source **15***j* starts to pass a current $I_{REF}=I_{11}$ according to the reference voltage V_{REF}'. In accordance with this, in the frequency/impedance conversion circuit **13**, a charge according to the current $I_{REF}=I_{11}$, is accumulated in the capacitive element **132**, and the voltage V_{REF} rises gradually. At this time, the oscillation signal is at the L level, the switch **133** is maintained in the off state, the switch **134** is maintained in the on state, the capacitive element **131** is in the discharge state, and the voltage V_{SC} thereof is substantially equal to the ground potential. Further, since the reference voltage V_{SENS} is maintained almost at the value V_{11}, the voltage difference detection circuit **16** gradually increases the control voltage V_{CTRL} as the voltage V_{REF} rises gradually.

(106) At timing t32, the voltage-controlled oscillation circuit **11** starts an oscillation operation, and the level of the oscillation signal becomes the H level. In accordance with this, the switch **133** is turned on, and the switch **134** is turned off. The voltage V_{REF} is lowered momentarily due to the fact that one end of the capacitive element **131** is connected to the input node **16***b*. After that, charge is redistributed between the capacitive elements **131** and **132**, and a charge according to the current $I_{REF}=I_{11}$ is injected into each of the capacitive elements **131** and **132**; accordingly, the voltage V_{SC} and the voltage V_{REF} rise gradually.

(107) At timing t33, the level of the oscillation signal becomes the L level. In accordance with this, the switch **133** is turned off, and the switch **134** is turned on. The capacitive element **131** is

discharged, and its voltage V.sub.SC is lowered to the ground potential. At this time, since the capacitive element **132** holds charge, the voltage V.sub.REF continues to rise gradually in accordance with a charge according to the current I.sub.REF=I.sub.11 being injected into the capacitive element **132**.

(108) At timing t34, the level of the oscillation signal becomes the H level. In accordance with this, the switch **133** is turned on, and the switch **134** is turned off. The voltage V.sub.REF is lowered momentarily due to the fact that one end of the capacitive element **131** is connected to the input node **16***b*. After that, charge is redistributed between the capacitive elements **131** and **132**, and a charge according to the current I.sub.REF=I.sub.11 is injected into each of the capacitive elements **131** and **132**; accordingly, the voltage V.sub.SC and the voltage V.sub.REF rise gradually.

(109) At timing t35, the level of the oscillation signal becomes the L level. In accordance with this, the switch **133** is turned off, and the switch **134** is turned on. The capacitive element **131** is discharged, and its voltage V.sub.SC is lowered to the ground potential. At this time, since the capacitive element **132** holds charge, the voltage V.sub.REF continues to rise gradually in accordance with a charge according to the current I.sub.REF=I.sub.11 being injected into the capacitive element **132**.

(110) At timings t36 to t44, similar operations to those at timings t34 and t35 and similar operations to those at timings t35 and t36 are alternately repeated; and the voltage V.sub.REF approaches the voltage V.sub.SENS=V.sub.11 on a time average basis while fluctuating in a sawtooth-like waveform, and the control voltage V.sub.CTRL gradually approaches the value V.sub.1. Accordingly, the frequency F.sub.OUT of the oscillation signal approaches a value F.sub.1 corresponding to the resistance value R.sub.SENS=R.sub.1 of the sensor **2**.

(111) When at timing t44 the voltage V.sub.REF becomes substantially equal to the reference voltage V.sub.SENS on a time average basis, then from timing t44 to timing t48, the control voltage V.sub.CTRL is stabilized at the value V.sub.1, the frequency F.sub.OUT of the oscillation signal is stabilized at a value F.sub.1 corresponding to the resistance value R.sub.SENS=R.sub.1 of the sensor **2**, and the frequency synchronization loop is set in a locked state. Thereby, the frequency synchronization circuit **10***j* stably outputs a frequency F.sub.OUT=F.sub.1 corresponding to the resistance value R.sub.SENS=R.sub.1 of the sensor **2**. The period TP1 is a period corresponding to the frequency F.sub.1.

(112) At timing t48, the resistance value R.sub.SENS of the sensor **2** changes to R.sub.2 (>R.sub.1) due to a state change or the like of the detection target of the sensor **2**, and the voltage changes to the voltage V.sub.SENS=V.sub.12 (<V.sub.11). In accordance with this, at timings t48 to t58, similar operations to those at timings t34 and t35 and similar operations to those at timings t35 and t36 are alternately repeated; and the voltage V.sub.REF approaches the reference voltage V.sub.SENS on a time average basis while fluctuating in a sawtooth-like waveform, and the control voltage V.sub.CTRL gradually approaches the value V.sub.3 (<V.sub.1). Accordingly, the frequency F.sub.OUT of the oscillation signal approaches a value F.sub.3 (>F.sub.1) corresponding to the resistance value R.sub.SENS=R.sub.2 of the sensor **2**.

(113) When at timing t58 the voltage V.sub.REF becomes substantially equal to the voltage V.sub.SENS on a time average basis, then on or after timing t58, the control voltage V.sub.CTRL is stabilized at the value V.sub.3, the frequency F.sub.OUT of the oscillation signal is stabilized at a value F.sub.3 corresponding to the resistance value R.sub.SENS=R.sub.2 of the sensor **2**, and the frequency synchronization loop is set in a locked state again. Thereby, the frequency synchronization circuit **10***j* stably outputs a frequency F.sub.OUT=F.sub.3 corresponding to the resistance value R.sub.SENS=R.sub.2 of the sensor **2**. The period TP3 is a period corresponding to the frequency F.sub.3.

(114) As above, in the second embodiment, in the sensor interface circuit **1***j*, the frequency synchronization circuit **10***j* is configured with a voltage change-type frequency synchronization circuit, and changes the voltage amplified and outputted by the amplification circuit **18***j* with

respect to the change in the detection value of the sensor **2**. Also with such a configuration, an oscillation frequency not depending on the accuracy of the reference voltage V.sub.REF′ can be obtained, and the accuracy of the operation of feedback of a frequency synchronization loop can be easily increased in the voltage domain.
(115) According to the present invention, an effect capable of reducing current consumption is exhibited.
(116) Although several embodiments according to the present invention have been described, these embodiments are presented for illustrative purposes only and are not intended to limit the scope of the invention. These novel embodiments can be implemented in various other forms, and various omissions, substitutions, and modifications can be made within the scope and spirit of the invention. The embodiments and modifications thereto are within the scope and spirit of the invention and are within the invention described in claims and equivalents thereof.

Claims

1. A sensor interface circuit comprising: a frequency synchronization circuit connectable to a sensor, the frequency synchronization circuit including: a reference voltage source configured to generate a reference voltage; a current source connected to the reference voltage source, the current source being configured to generate a current by using the reference voltage; a voltage difference detection circuit including a first input node connected to the reference voltage source, a second input node connected to the current source, and an output node, one of a voltage received at the first input node and a voltage received at the second input node corresponding to a detection value of the sensor, and the voltage difference detection circuit being configured to generate a control voltage according to a difference between the voltage received at the first input node and the voltage received at the second input node; a voltage-controlled oscillation circuit connected to the output node of the voltage difference detection circuit, the voltage-controlled oscillation circuit being configured to generate an oscillation signal according to the control voltage; and a frequency/impedance conversion circuit connected between the voltage-controlled oscillation circuit and the second input node of the voltage difference detection circuit, the frequency/impedance conversion circuit being configured to convert a frequency of a signal according to the oscillation signal into impedance.
2. The sensor interface circuit according to claim 1, wherein the current source is connectable to the sensor and is configured to generate a current by using the reference voltage and the detection value of the sensor, and in the voltage difference detection circuit, the voltage received at the second input node corresponds to the detection value of the sensor and the frequency.
3. The sensor interface circuit according to claim 2, wherein the current source includes: a first transistor connected between a second potential and the second input node of the voltage difference detection circuit; a second transistor connectable between the second potential and the sensor; and a differential amplification circuit including a first input node connected to the reference voltage source, a second input node connected to a node between the second transistor and the sensor, and an output node connected in common to a gate of the first transistor and a gate of the second transistor.
4. The sensor interface circuit according to claim 1, further comprising: an amplification circuit connected between the reference voltage source and the voltage difference detection circuit and connectable to the sensor, the amplification circuit being configured to amplify a difference between the reference voltage and the detection value of the sensor and output a voltage according to the amplified difference, wherein in the voltage difference detection circuit, the voltage received at the first input node corresponds to the detection value of the sensor, and the voltage received at the second input node corresponds to the frequency.
5. The sensor interface circuit according to claim 4, wherein the current source includes: a resistive

element; a first transistor connected between a second potential and the second input node of the voltage difference detection circuit; a second transistor connected between the second potential and the resistive element; and a differential amplification circuit including a first input node connected to the amplification circuit, a second input node connected to a node between the second transistor and the resistive element, and an output node connected in common to a gate of the first transistor and a gate of the second transistor.

6. The sensor interface circuit according to claim 1, further comprising: a frequency division circuit connected between the voltage-controlled oscillation circuit and the frequency/impedance conversion circuit, the frequency division circuit being configured to perform frequency division on the oscillation signal, wherein the frequency/impedance conversion circuit is configured to convert a frequency of the signal subjected to frequency division into impedance.

7. The sensor interface circuit according to claim 1, wherein the frequency/impedance conversion circuit is configured with a switched capacitor circuit.

8. The sensor interface circuit according to claim 7, wherein the frequency/impedance conversion circuit includes: a first capacitive element of which one end is connected to a first potential; a second capacitive element of which one end is connected to the first potential and another end is connected to the second input node of the voltage difference detection circuit; a first switch of which one end is connected to the second input node of the voltage difference detection circuit, another end is connected to another end of the first capacitive element, and a control end is connected to the voltage-controlled oscillation circuit; an inverter connected to the voltage-controlled oscillation circuit; and a second switch of which one end is connected to the other end of the first capacitive element, another end is connected to the first potential, and a control end is connected to the inverter.

9. A sensor module comprising: an antenna; an impedance conversion circuit; a sensor; and the sensor interface circuit according to claim 1, the sensor interface circuit being connected between the sensor and the antenna.